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NEM 358 – Final Project

Single Phase CFD Simulation of a VVER-1200 Fuel Rod Bundle

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1 Introduction

The VVER-1200 (V-392M) is a pressurized water reactor rated at 3200 MW thermal, with a core built from 163 hexagonal fuel assemblies, 312 rods each, arranged around a central guide channel [1]. The goal of this project is to build a single-phase, steady state CFD model of a small representative piece of one of these assemblies and look at the coolant temperature rise, the pressure drop, and the maximum fuel temperature, then check these numbers against simple analytical estimates [4].

Only a short axial slice is modeled – about one tenth of the full active fuel height (37.5 cm out of 375 cm). Since the axial power shape in a core is roughly sinusoidal and peaks near the mid-plane, this slice represents the bottom part of the core where the local power is still ramping up, so a sinusoidal volumetric heat source that starts from (almost) zero at the inlet was used as the boundary condition.

Conjugate heat transfer (CHT) is used throughout – the fuel rods and the central guide tube are solid bodies with their own conductivities, and the heat generated inside them conducts out to the rod surface before being picked up by the coolant. This is the only way to get a meaningful fuel centerline temperature out of a CFD run, since a pure-fluid model with a wall heat flux BC wouldn't tell us anything about the temperature distribution inside the rods.

The simulation was carried out in **Simcenter STAR-CCM+** [3]. Section 2 covers the geometry, mesh, boundary conditions and the analytical correlations used for comparison. Section 3 presents and discusses the results. Section 4 gives the conclusions, and the appendix collects the more detailed input data.

2 Methodology

2.1 Geometry

Table 1 lists the reactor and assembly data from the project handout [1].

Table 1: Reactor and fuel assembly design parameters.

Parameter	Value
Thermal power, P_{th}	3200 MW
Number of fuel assemblies	163
Rods per assembly	312
Active fuel height, H_f	3.75 m
Axial peaking factor, K_z	1.53
Fuel pellet diameter	7.6 mm
Fuel rod outer diameter, D_{rod}	9.1 mm
Guide tube outer diameter, D_{guide}	12.9 mm
Rod pitch, P	12.75 mm
Core inlet temperature	298.2 °C (571.35 K)
Core average outlet temperature	328.9 °C
Property evaluation state	15.51 MPa / 309.4 °C

The domain is a 19-rod hexagonal mini-bundle: 18 fuel rods + 1 central guide tube, pitch $P = 12.75$ mm. Power factors: 12 outer-ring rods $f = 0.97$, 6 inner-ring rods $f = 0.88$, guide tube $f = 0$. Axial length $L = 0.375$ m, $z \in [0, 0.375]$ m, $z = 0 = \text{inlet}$.

$$R_{hex} = 2.5754 P = 2.5754 \times 12.75 = 32.836 \text{ mm} \quad (1)$$

$$P_{wet} = 18\pi D_{rod} + \pi D_{guide} + 6R_{hex} = 0.5146 + 0.04053 + 0.19702 = 0.7521 \text{ m} \quad (2)$$

$$A_{hex} = \frac{3\sqrt{3}}{2} R_{hex}^2 = 2.801 \times 10^{-3} \text{ m}^2 \quad (3)$$

$$A_{rods} = 18 \cdot \frac{\pi}{4} D_{rod}^2 = 1.171 \times 10^{-3} \text{ m}^2 \quad (4)$$

$$A_{guide} = \frac{\pi}{4} D_{guide}^2 = 1.307 \times 10^{-4} \text{ m}^2 \quad (5)$$

$$A_{flow} = A_{hex} - A_{rods} - A_{guide} = 1.500 \times 10^{-3} \text{ m}^2 \quad (6)$$

$$D_h = \frac{4A_{flow}}{P_{wet}} = 7.974 \text{ mm} \quad (7)$$

2.2 Mesh

Polyhedral core mesh with prism layers on every solid surface (18 rod walls, guide tube wall and outer hexagon wall). 19 fluid-solid contact interfaces were generated automatically via the imprint operation.

- Base size: 2 mm
- Prism layers: 6, total thickness 1 mm, stretch factor 1.3
- Resulting volume mesh: $\approx 1.7\text{--}1.8 \times 10^6$ cells

Simcenter STAR-CCM+

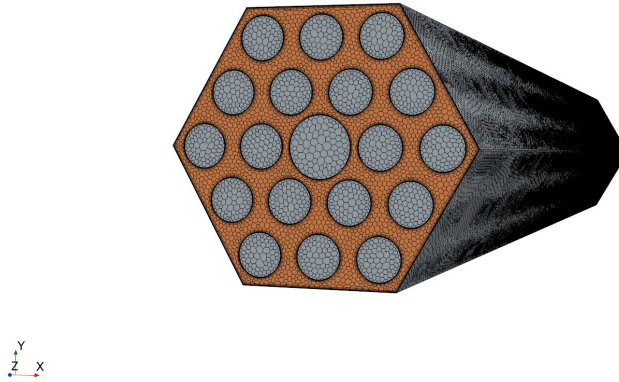


Figure 1: Volume mesh of the 19-rod hexagonal mini-bundle (polyhedral cells with prism layers near the rod and guide-tube surfaces).

2.3 Physics Models and Boundary Conditions

Steady, segregated flow + segregated energy, SST $k\text{-}\omega$ with $\gamma\text{-}Re_\theta$ transition [3], constant-density liquid (Richardson number $\sim 10^{-4}$), gravity $(0, 0, -9.81) \text{ m/s}^2$, reference pressure 15.51 MPa.

- **Inlet:** velocity inlet, $V = 5.842 \text{ m/s}$ ($+z$), $T = 571.35 \text{ K}$, turbulence intensity 4%.

- **Outlet:** pressure outlet, $P_{\text{gauge}} = 0 \text{ Pa}$.
- **Outer hexagon wall:** adiabatic.
- **Rod / guide tube surfaces:** coupled CHT walls; roughness $k_s = 1 \times 10^{-6} \text{ m}$ (fuel rods) and $k_s = 2 \times 10^{-6} \text{ m}$ (guide tube).

2.4 Material Properties

Table 2: Coolant properties at 15.51 MPa, 309.4 °C.

Property	Value
ρ	706.157 kg/m ³
c_p	5723.46 J/(kg K)
k	0.541 66 W/(m K)
μ	$8.48 \times 10^{-5} \text{ Pa s}$

Table 3: Effective solid properties.

Property	Fuel rod (effective)	Guide tube (steel)
k	4.8 W/(m K)	15.2 W/(m K)
c	330 J/(kg K)	482 J/(kg K)
ρ	10 450 kg/m ³	8065 kg/m ³

2.5 Volumetric Heat Generation

$$N_{\text{rods}} = 163 \times 312 = 50856 \quad (8)$$

$$V_{\text{fuel}} = N_{\text{rods}} \cdot \frac{\pi}{4} D_{\text{pellet}}^2 H_f = 8.652 \text{ m}^3 \quad (9)$$

$$\bar{q}''' = \frac{P_{th}}{V_{\text{fuel}}} = 3.698 \times 10^8 \text{ W/m}^3 \quad (10)$$

Axial shape $q'''(z) \propto \sin(\pi z/H_f)$, normalized by $\pi/2$:

$$q'''_{\text{outer}}(z) = \bar{q}''' \cdot 0.97 \cdot \frac{\pi}{2} \cdot \sin\left(\frac{\pi z}{3.75}\right) = 5.6346 \times 10^8 \sin\left(\frac{\pi z}{3.75}\right) \text{ W/m}^3 \quad (11)$$

$$q'''_{\text{inner}}(z) = \bar{q}''' \cdot 0.88 \cdot \frac{\pi}{2} \cdot \sin\left(\frac{\pi z}{3.75}\right) = 5.1118 \times 10^8 \sin\left(\frac{\pi z}{3.75}\right) \text{ W/m}^3 \quad (12)$$

Radial split: $\sqrt{x^2 + y^2} > 18 \text{ mm} \Rightarrow$ outer coefficient, else inner. Guide tube: $q''' = 0$.

2.6 Analytical Reference Solutions

Coolant temperature rise (energy balance)

$$Q_{total} \approx 12 \times 2141 \text{ W} + 6 \times 1942 \text{ W} \approx 37.3 \text{ kW} \quad (13)$$

$$\dot{m} = \rho V A_{flow} = 6.193 \text{ kg/s} \quad (14)$$

$$\Delta T_{energy} = \frac{Q_{total}}{\dot{m} c_p} = 1.053 \text{ K} \quad (15)$$

Reynolds number

$$Re = \frac{\rho V D_h}{\mu} = \frac{706.157 \times 5.842 \times 0.007974}{8.48 \times 10^{-5}} \approx 3.88 \times 10^5 \quad (16)$$

Surface roughness check $k_s/D_h \approx 1.25 \times 10^{-4}$ – hydraulically smooth; roughness terms omitted from the friction factor.

Pressure drop (Rehme correlation [2])

$$f = \frac{a + b_1(P/D_h - 1) + b_2(P/D_h - 1)^2}{Re^{0.18}}, \quad a = 0.1339, \quad b_1 = 0.09059, \quad b_2 = -0.09926 \quad (17)$$

$$P/D_h - 1 = \frac{12.75}{7.974} - 1 = 0.5995 \quad (18)$$

$$f = \frac{0.1339 + 0.09059(0.5995) - 0.09926(0.5995)^2}{(3.88 \times 10^5)^{0.18}} = \frac{0.1525}{10.83} = 0.01408 \quad (19)$$

$$\Delta P_{hydro} = \rho g L = 706.157 \times 9.80665 \times 0.375 = 2596.9 \text{ Pa} \quad (20)$$

$$\Delta P_{friction} = f \cdot \frac{L}{D_h} \cdot \frac{\rho V^2}{2} = 0.01408 \times 47.03 \times 12064.5 = 7993.8 \text{ Pa} \quad (21)$$

$$\Delta P_{total} = \Delta P_{hydro} + \Delta P_{friction} = 10\,590.7 \text{ Pa} \quad (22)$$

Peak fuel temperature (1-D conduction)

$$Nu = 0.023 Re^{0.8} Pr^{0.4}, \quad Pr = 0.896 \Rightarrow Nu \approx 653, \quad h \approx 44\,360 \text{ W/(m}^2 \text{ K)} \quad (23)$$

$$q'''_{max} = 5.6346 \times 10^8 \times \sin(0.1\pi) = 1.741 \times 10^8 \text{ W/m}^3 \quad (24)$$

$$q'' = \frac{q'''_{max} R}{2} \approx 3.96 \times 10^5 \text{ W/m}^2, \quad \Delta T_{film} = \frac{q''}{h} \approx 8.9 \text{ K} \quad (25)$$

$$\Delta T_{radial} = \frac{q'''_{max} R^2}{4k_{fuel}}, \quad R = 4.55 \text{ mm} \Rightarrow \Delta T_{radial} \approx 187.7 \text{ K} \quad (26)$$

$$T_{max} = T_{coolant,out} + \Delta T_{film} + \Delta T_{radial} \quad (27)$$

3 Results and Discussion

The run converged in 358 iterations with all residuals (continuity, momentum, turbulence, transition variables and energy) dropping below 10^{-3} . Table 4 summarises the key integrated output.

Table 4: Summary of CFD results from the converged Star-CCM+ simulation.

Quantity	Value
Inlet mass-flow-averaged temperature	571.350 K
Outlet mass-flow-averaged temperature	572.383 K
Coolant temperature rise, ΔT	1.033 K
Maximum fuel rod temperature, T_{max}	770.405 K
Total pressure drop, ΔP	9728.6 Pa

3.1 Temperature

Figure 2 shows the outlet temperature contour. The coolant sits mostly at 572 K, with a thin warmer ring around each fuel rod surface. The central guide tube doesn't show this ring since it generates no heat.

Simcenter STAR-CCM+

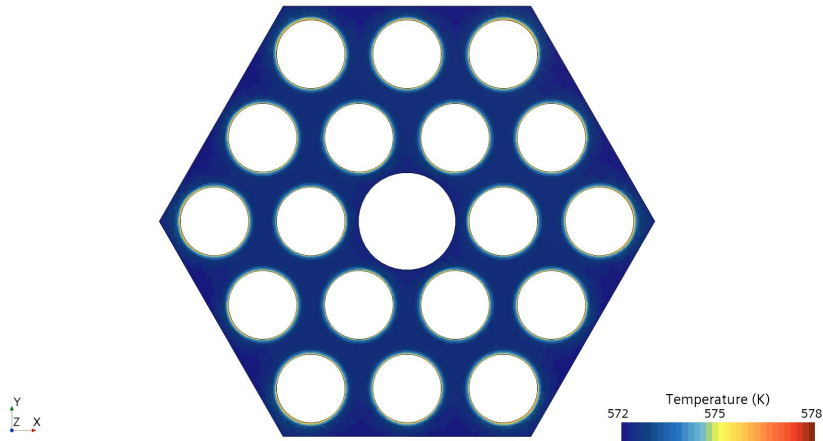


Figure 2: Temperature contour at the outlet (572 K to 578 K).

The diagonal probe (Figure 3) passes through both coolant gaps and solid rods. The flat region around 572–575 K is the coolant, while the spikes reaching up to 770 K are the solid rod cross-sections – peak values near the rod centrelines, consistent with the T_{max} in Table 4.

Figure 4 shows the axial coolant temperature along a line probe from inlet ($z = 0$) to outlet ($z = 0.375$ m). The profile is almost flat near $z = 0$ where $q''' \rightarrow 0$ and the slope picks up toward the outlet – exactly what the sinusoidal heat source produces over this short slice.

The coolant temperature rise $\Delta T = 1.033$ K is within 1.9% of the energy-balance estimate of 1.053 K, confirming that energy is being conserved properly in the simulation.

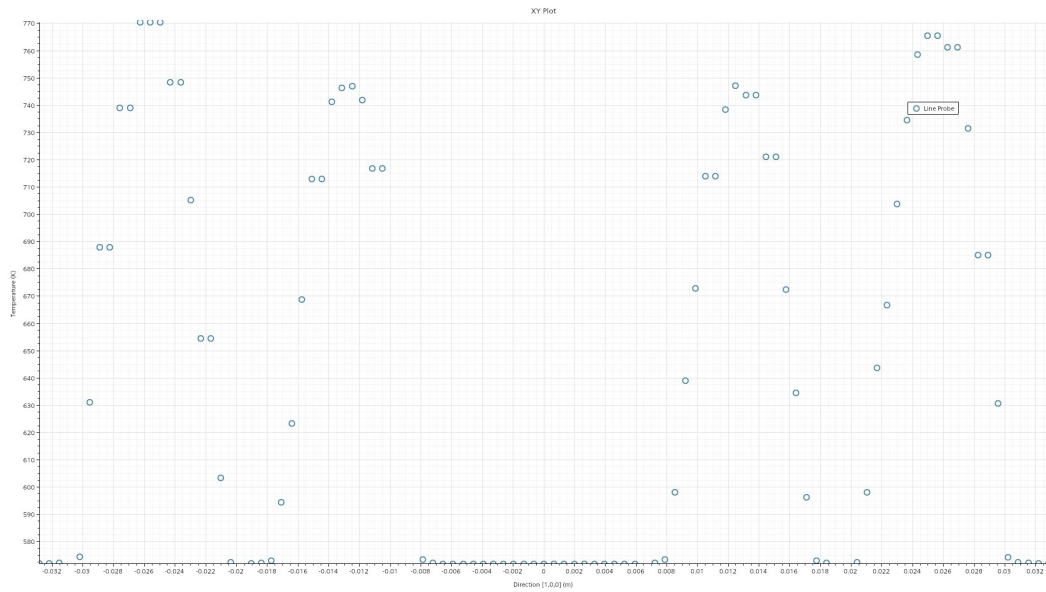


Figure 3: Temperature along the diagonal outlet probe. Flat region = coolant gaps; spikes = solid rod cross-sections.

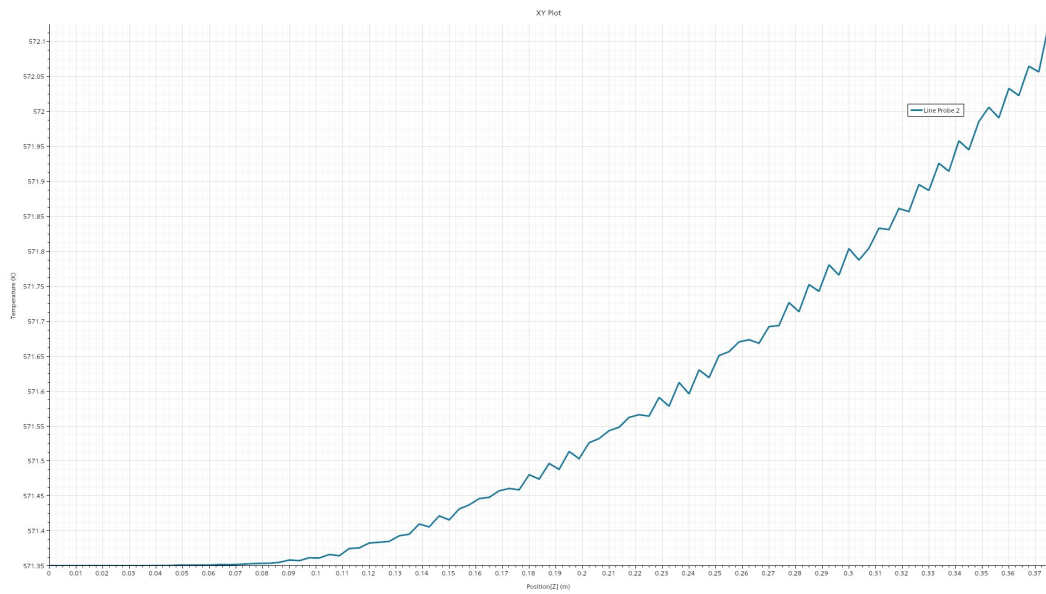


Figure 4: Coolant temperature along an axial line from inlet to outlet.

3.2 Pressure

Figure 5 shows the outlet pressure contour. The 7.6–9.5 kPa range includes local dynamic pressure variation around each rod and should not be read as the gauge static pressure. The diagonal probe (Figure 6) confirms this: the actual gauge pressure at the outlet stays within -10 to -230 Pa – essentially zero, as the pressure-outlet boundary condition requires.

Simcenter STAR-CCM+

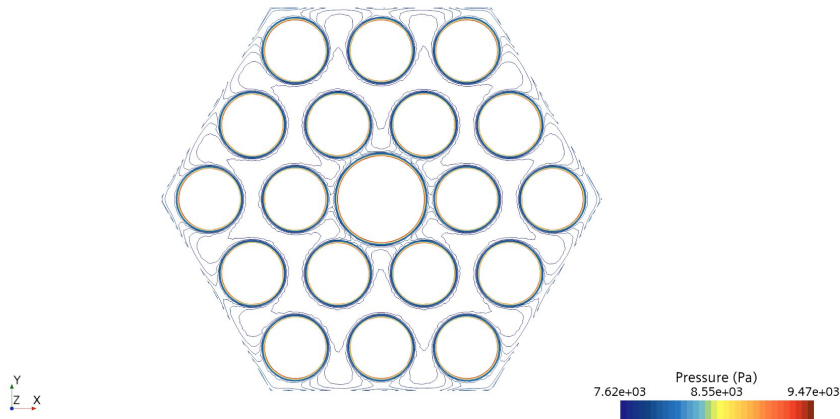


Figure 5: Pressure contour at the outlet.

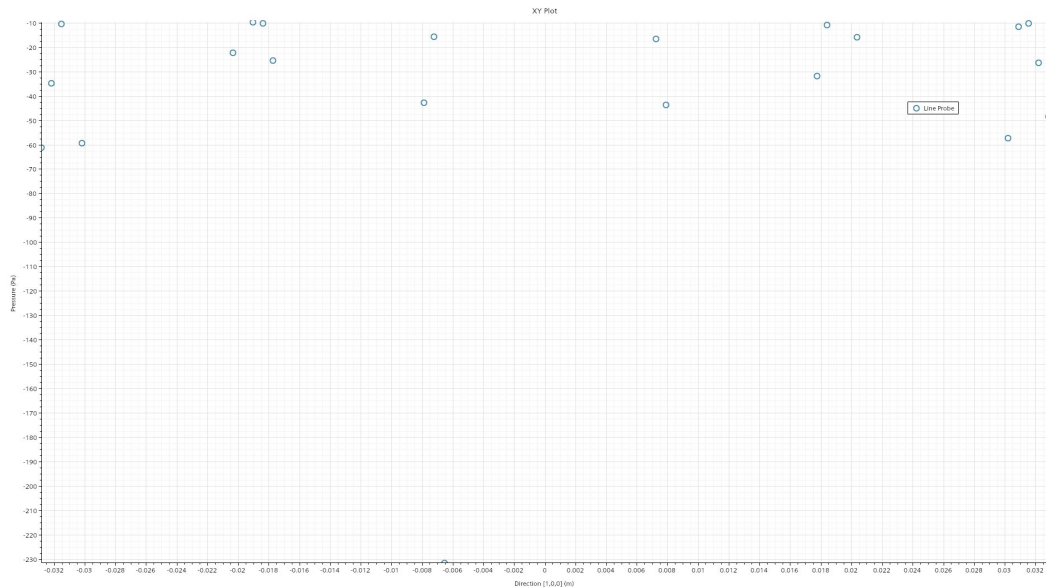


Figure 6: Gauge pressure along the diagonal outlet probe – near zero, confirming the outlet boundary condition.

The total pressure drop is 9728.6 Pa, compared to the Rehme analytical estimate of 10 590.7 Pa – a difference of 8.1%. Both land on the same side (CFD below the correlation), which is not unusual since Rehme’s formula was derived for infinite rod arrays and tends to slightly over-predict for a small, finite bundle with an outer wall.

3.3 Velocity

The outlet velocity contour (Figure 7) is fairly uniform (5.69–6.05 m/s), consistent with the 5.842 m/s inlet velocity. The diagonal probe (Figure 8) shows the expected pattern for rod-bundle flow: velocity peaks in the narrow gaps between rods (up to ~ 6.5 m/s) and drops near the walls (down to ~ 2.3 m/s). The periodicity follows the lattice spacing.

Simcenter STAR-CCM+

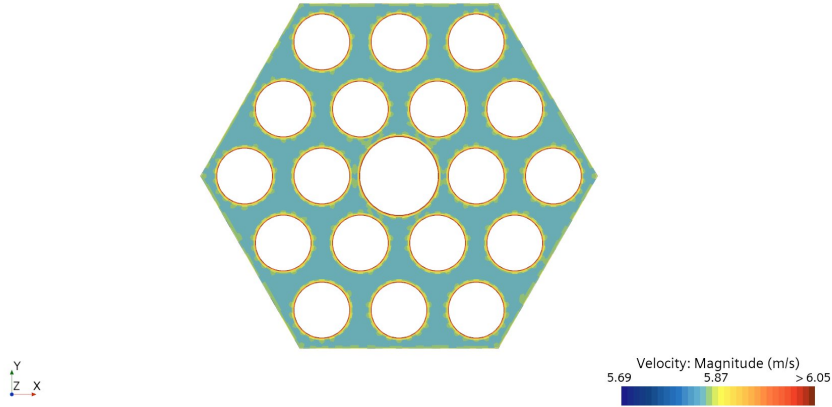


Figure 7: Velocity magnitude contour at the outlet.

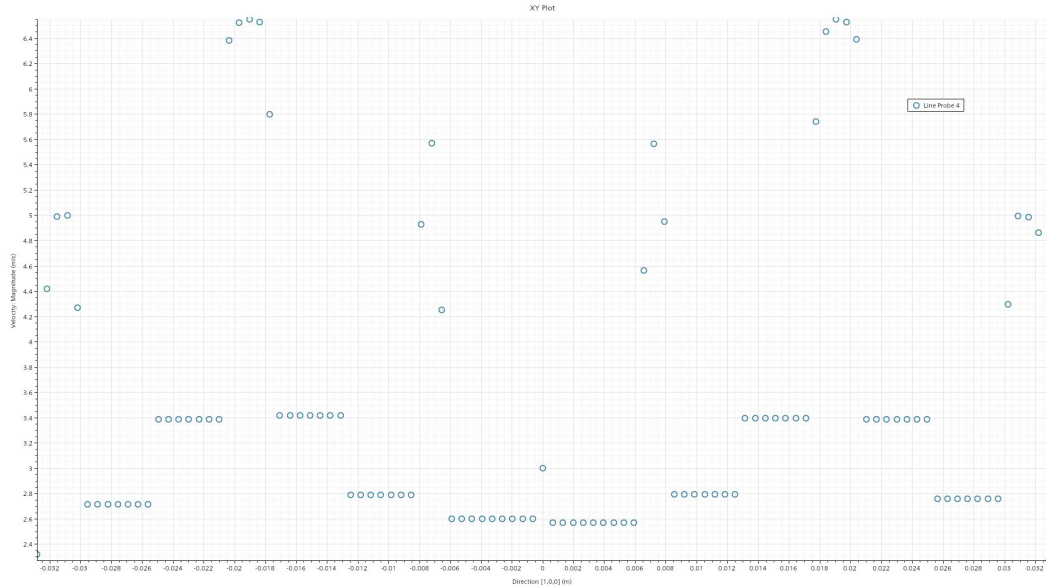


Figure 8: Velocity magnitude along the diagonal outlet probe.

3.4 Comparison with Analytical Solutions

For the analytical T_{max} , the rod surface temperature is estimated as the CFD outlet coolant temperature plus the film rise ($\Delta T_{film} \approx 8.9$ K), giving ≈ 581.3 K, and adding $\Delta T_{radial} = 187.7$ K gives $T_{max,analytical} \approx 769.0$ K.

All three come in within about 9%, which is a solid result for this kind of correlation-vs-CFD

Table 5: CFD results compared against analytical estimates.

Quantity	CFD (Star-CCM+)	Analytical reference	Difference
ΔP [Pa]	9728.6	10590.7 (Rehme)	8.1%
ΔT [K]	1.033	1.053 (energy balance)	1.9%
T_{max} [K]	770.4	769.0 (1-D conduction)	0.18%

comparison. The pressure drop gap (8.1%) is the largest and the most explainable – Rehme doesn’t account for the actual geometry of a finite hexagonal bundle with an outer wall and a guide tube at the centre. The temperature rise match (1.9%) is basically a conservation check. The peak fuel temperature match (0.18%) is almost suspiciously good, but it makes sense here because the outer-ring rods are cooled around their *entire* circumference in this full-hexagon geometry, which is exactly the assumption the 1-D cylindrical conduction formula makes.

4 Conclusions

A conjugate heat transfer CFD model of a 19-rod (18 fuel + 1 guide tube) hexagonal mini-bundle, covering the bottom 37.5 cm (1/10) of the active core height, was built and solved in Star-CCM+ with a sinusoidal axial heat source that starts from near zero at the inlet. Going with the full 19-pin hexagon instead of a symmetry sector turned out to be a good call – every rod is cooled around its whole circumference, which made the analytical comparisons clean and straightforward.

The CFD numbers came out close to the analytical estimates across the board: 8.1% on the pressure drop, 1.9% on the temperature rise, and 0.18% on the peak fuel temperature. The peak fuel temperature (770.4 K) is far below any safety-relevant limit, which makes sense given that this slice sits near the bottom of the core where the power is still ramping up. The velocity and pressure fields at the outlet both look physically sensible – faster flow in the gaps, near-zero gauge pressure variation – which gives confidence that the mesh and solver settings are doing the right thing.

With more time, a mesh refinement study would be the obvious next step (only about 2 cells fit across the narrowest rod-to-rod gap right now), along with extending the model to a slice nearer the core mid-plane where the power is actually at its peak and the analytical margins get tighter. The optional transient loss-of-flow case wasn't attempted within the scope of this project.

5 References

References

- [1] Hacettepe University, Department of Nuclear Engineering. NEM 358 Final Project Description and VVER-1200 Design Data Sheet, 2026.
- [2] K. Rehme. Pressure drop correlations for fuel element spacers. *Nuclear Technology*, 17, 1973.
- [3] Siemens Digital Industries Software. *Simcenter STAR-CCM+ User Guide*. Siemens, 2502 (20.02.007) edition.
- [4] Neil E. Todreas and Mujid S. Kazimi. *Nuclear Systems Volume I: Thermal Hydraulic Fundamentals*. CRC Press, 2nd edition, 2011.

Appendix

A. Summary of Input Parameters

Parameter	Value
P	12.75 mm
D_{rod}	9.1 mm
D_{guide}	12.9 mm
R_{hex}	32.836 mm
A_{flow}	$1.500 \times 10^{-3} \text{ m}^2$
P_{wet}	0.7521 m
D_h	7.974 mm
L	0.375 m
V_{inlet}	5.842 m/s
Re	3.88×10^5
$\dot{m}$	6.193 kg/s

Table 6: Consolidated geometric and flow parameters.

B. Heat Generation Field Functions

Outer rods ($r > 18 \text{ mm}$, $f = 0.97$):

$$5.6346\text{e}8 * \sin(3.14159265359 * \text{\$Position}[2] / 3.75)$$

Inner rods ($r \leq 18 \text{ mm}$, $f = 0.88$):

$$5.1118\text{e}8 * \sin(3.14159265359 * \text{\$Position}[2] / 3.75)$$

Radial check:

$$\sqrt{(\text{\$Position}[0] * \text{\$Position}[0]) + (\text{\$Position}[1] * \text{\$Position}[1])} > 0.018$$

Central guide tube: $q''' = 0$ (no heat source assigned).

C. Raw Report Output from Star-CCM+

Inlet mass-flow-averaged T = 571.350 K
 Outlet mass-flow-averaged T = 572.383 K
 Maximum T (Fuel_Rods_Solid) = 770.405 K
 Total pressure drop = 9728.552 Pa